



Examination Number:

Set:

**Shore**

**Year 12**

**HSC Assessment Task 5 - Trial HSC**

**14th August 2015**

# **Mathematics Extension 1**

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## **General Instructions**

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this question paper
- Answer Questions 1–10 on the Multiple Choice Answer Sheet provided
- Start each of Questions 11–14 in a new writing booklet
- In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.
- Write your examination number on the front cover of each booklet
- If you do not attempt a question, submit a blank booklet marked your examination number and “N/A” on the front cover

**Total marks – 70**

**Section I** Pages 3–6

**10 marks**

- Attempt Questions 1–10
- Allow about 15 minutes for this section

**Section II** Pages 7–12

**60 marks**

- Attempt Questions 11–14
- Allow about 1 hour 45 minutes for this section

**Note: Any time you have remaining should be spent revising your answers.**

**DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM**

## Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the Multiple Choice Answer Sheet for Questions 1–10.

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1 What is the value of  $\int_0^{\frac{1}{5}} \frac{1}{\sqrt{1-25x^2}} dx$ ?

- (A)  $\frac{\pi}{10}$
- (B)  $\frac{\pi}{5}$
- (C)  $\frac{\pi}{2}$
- (D)  $\frac{\pi}{25}$

2 Which of the following is a simplification of  $4\log_e \sqrt{e^x}$ ?

- (A)  $4\sqrt{x}$
- (B)  $\frac{1}{2}x$
- (C)  $2x$
- (D)  $x^2$

3 The acute angle between the lines  $2x - y = 0$  and  $kx - y = 0$  is equal to  $\frac{\pi}{4}$ .  
What is the value of  $k$ ?

- (A)  $k = -3$  or  $k = -\frac{1}{3}$
- (B)  $k = -3$  or  $k = \frac{1}{3}$
- (C)  $k = 3$  or  $k = -\frac{1}{3}$
- (D)  $k = 3$  or  $k = \frac{1}{3}$

4 What is the domain and range of  $y = 2\cos^{-1}(x - 1)$ ?

- (A) Domain :  $0 \leq x \leq 2$ . Range:  $0 \leq y \leq \pi$
- (B) Domain :  $-1 \leq x \leq 1$ . Range:  $0 \leq y \leq \pi$
- (C) Domain :  $0 \leq x \leq 2$ . Range:  $0 \leq y \leq 2\pi$
- (D) Domain :  $-1 \leq x \leq 1$ . Range:  $0 \leq y \leq 2\pi$

5 Which of the following is a simplification of  $\frac{1 - \cos 2x}{\sin 2x}$ ?

- (A)  $1 - \cot x$
- (B)  $1$
- (C)  $\cot x$
- (D)  $\tan x$

6 The domain of the function  $y = x^2(x - 2)^2$  must be restricted to have an inverse function.

Which of the following restrictions on the domain allows for an inverse function to exist?

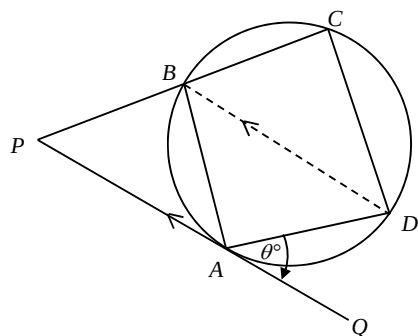
- (A)  $x \geq 2$
- (B)  $0 \leq x \leq 2$
- (C)  $x \geq 0$
- (D)  $x \leq 2$

- 7 A particle is moving in simple harmonic motion with displacement  $x$ .  
The velocity  $v$  of the particle is given by

$$v^2 = 4(25 - x^2).$$

What is the amplitude  $a$  of the motion and the maximum speed of the particle?

- (A)  $a = 2$  and maximum speed is 5 m/s  
(B)  $a = 2$  and maximum speed is 10 m/s  
(C)  $a = 5$  and maximum speed is 5 m/s  
(D)  $a = 5$  and maximum speed is 10 m/s
- 8 In the diagram  $ABCD$  is a cyclic quadrilateral. The tangent  $PQ$  touches the circle at  $A$ . The diagonal  $BD$  is parallel to the tangent  $PQ$ .  $QA$  produced intersects with  $CB$  at  $P$ . Let  $\angle QAD = \theta^\circ$



NOT  
TO  
SCALE

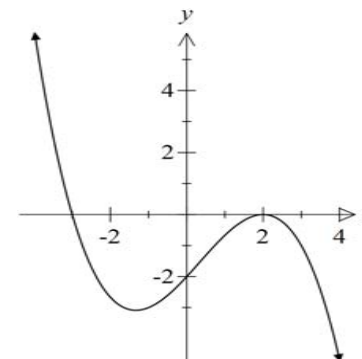
What is the size of  $\angle BCD$ ?

- (A)  $\theta^\circ$   
(B)  $2\theta^\circ$   
(C)  $(\pi - \theta)^\circ$   
(D)  $(\pi - 2\theta)^\circ$

- 9 Which of the following is the general solution of  $2\sin \frac{x}{2} = \sqrt{3}$ ?

- (A)  $x = 4n\pi \pm \frac{\pi}{3}$ , where  $n$  is an integer.  
(B)  $x = 2n\pi + \frac{\pi}{3}$ , where  $n$  is an integer.  
(C)  $x = 2n\pi + (-1)^n \times \frac{2\pi}{3}$ , where  $n$  is an integer.  
(D)  $x = 4n\pi + (-1)^n \times \frac{2\pi}{3}$ , where  $n$  is an integer.

- 10 The polynomial graph shown below has equation  $y = A(x + B)(x + C)^2$ .



Which of the following is true?

- (A)  $A = -\frac{1}{6}$ ,  $B = 3$ ,  $C = -2$   
(B)  $A = \frac{1}{6}$ ,  $B = -3$ ,  $C = 2$   
(C)  $A = 1$ ,  $B = 3$ ,  $C = -2$   
(D)  $A = -1$ ,  $B = 3$ ,  $C = -2$

**End of Section I**

## Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

**Question 11** (15 marks) Use a SEPARATE writing booklet

(a) Solve the inequality  $\frac{x-3}{x+4} \leq 2$ . 3

(b) The point  $C(-3, 8)$  divides the interval  $AB$  externally in the ratio  $k : 1$ .  
Find the value of  $k$  if  $A$  is the point  $(6, -4)$  and  $B$  is the point  $(0, 4)$ . 2

(c) Find  $\frac{d}{dx}(x \sin^{-1} 2x)$ . 2

(d) Use the substitution  $u = \ln x$  to evaluate  $\int_{\frac{1}{e}}^{\sqrt{e}} \frac{dx}{x\sqrt{1 - (\ln x)^2}}$ . 3

(e) The region bounded by the curve  $y = \cos 3x$  and the  $x$  axis between the lines  $x = 0$  and  $x = \frac{\pi}{6}$  is rotated through one complete revolution about the  $x$ -axis.  
Find the exact volume of the solid formed. 3

(f) Use one application of Newton's method with an initial approximation of  $x = 1.5$  to find the next approximation to the root of the equation  $2 \log_e x - \cos x = 0$ . Give your answer correct to 2 significant figures. 2

**Question 12** (15 marks) Use a SEPARATE writing booklet

(a) Find the constant term of the expansion  $\left(\frac{1}{x^2} - 2x^3\right)^{10}$ . 3

(b) The polynomial  $P(x) = x^3 + ax^2 + bx - 6$  has a remainder of 8 when divided by  $(x+1)$  and  $(x-3)$  is a factor of the polynomial  $P(x)$ . Find the values of  $a$  and  $b$ . 3

(c) The displacement  $x$  metres of a particle moving in simple harmonic motion is given by

$$x = 4 \cos \pi t$$

where time  $t$  is in seconds.

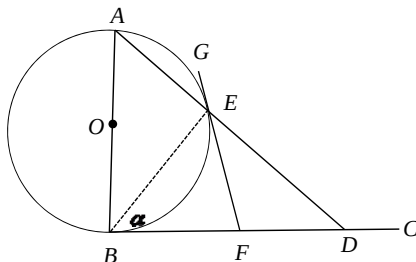
(i) What is the period of the oscillation? 1

(ii) What is the speed of the particle as it moves through the equilibrium position? 1

**Question 12 continues on the following page**

Question 12 (continued)

- (d) In the diagram  $AB$  is a diameter of the circle centre,  $O$ , and  $BC$  is tangential to the circle at  $B$ . The line  $AED$  intersects the circle at  $E$  and  $BC$  at  $D$ . The tangent to the circle at  $E$  intersects  $BC$  at  $F$ . Let  $\angle EBF = \alpha$ .



Copy or trace the diagram into your writing booklet.

- (i) Prove that  $\angle FED = \frac{\pi}{2} - \alpha$ . 2
- (ii) Prove that  $BF = FD$ . 2
- (e) Use mathematical induction to prove that  $5^n + 2 \times 11^n$  is a multiple of 3 for all integers  $n \geq 1$ . 3

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

- (a) An oven which had been heated to  $180^\circ\text{C}$  was switched off when the cook was finished baking at 11:30 am. The oven was in a kitchen which was kept at a constant temperature of  $22^\circ\text{C}$ .

After  $t$  minutes, the temperature,  $T^\circ\text{C}$ , of the oven was given by:

$$T = 22 + Be^{-kt}$$

where  $A$ ,  $B$  and  $k$  are positive constants.

- (i) If after 10 minutes, the oven's temperature has dropped to  $115^\circ\text{C}$ , find  $B$  and the exact value of  $k$ . 2
- (ii) At what time will the oven's temperature drop to  $23^\circ\text{C}$ . 2  
Give your answer correct to the nearest minute.

- (b) A particle is moving along the  $x$ -axis so that its acceleration after  $t$  seconds is given by

$$\ddot{x} = -e^{-\frac{x}{2}}$$

The particle starts at the origin with an initial velocity of 2 m/s.

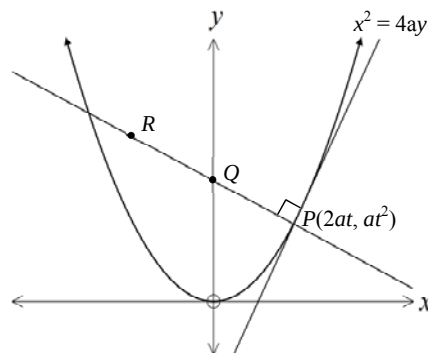
- (i) If  $v$  is the velocity of the particle, find  $v^2$  as a function of  $x$ . 2
- (ii) Given that  $v > 0$  throughout the motion, show that the displacement  $x$  as a function of time  $t$  is given by 3

$$x = 4 \log_e \left( \frac{t+2}{2} \right)$$

Question 13 continues on the following page

Question 13 (continued)

(c)



The diagram shows the parabola  $x^2 = 4ay$ . The normal at the point  $P(2at, at^2)$  cuts the  $y$ -axis at  $Q$  and is produced to a point  $R$  on the normal such that  $PQ = QR$ .

- (i) Show that the equation of the normal at  $P$  is given by 2

$$x + ty = 2at + at^3.$$

- (ii) Find the coordinates of  $Q$  and show that the coordinates of  $R$  are  $(-2at, at^2 + 4a)$ . 2

- (iii) Show that the locus of  $R$  is another parabola and state its vertex. 2

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) A particle is projected under gravity  $g$  with speed  $V$  metres per second at an angle of  $\theta$  from a point  $O$  on horizontal ground. It strikes the ground at  $P$ , where  $OP = R$ .

The equations of motion of the particle are

$$x = Vt \cos \theta \quad \text{and} \\ y = Vt \sin \theta - \frac{1}{2}gt^2 \quad (\text{Do not derive these equations})$$

- (i) If  $\theta$  is  $45^\circ$  show that the equation of trajectory of the particle is given by 2

$$y = x - g \frac{x^2}{V^2}$$

- (ii) Hence, or otherwise, show that  $R = \frac{V^2}{g}$ . 2

- (iii) A bullet is fired from  $O$  with velocity 30 m/s at an angle of  $45^\circ$  to the horizontal. Find the speed of the ball when it has travelled a horizontal distance of 15 m from its starting point. (Take  $g = 10 \text{ ms}^{-2}$ ). Give your answer correct to 1 decimal place. 3

- (b) Initially, a ladder leaning against a wall just reaches a window sill 2.4 metres above the ground. The foot of the ladder is  $x$  metres from the wall and it makes an angle of  $\theta$  (in radians) with the horizontal. The foot of the ladder is slipping at a rate of 2 cm/s.

- (i) Show that  $\frac{dx}{d\theta} = -\frac{2.4}{\sin^2 \theta}$ . 2

- (ii) Find the rate of change of the angle when  $\theta = \frac{\pi}{4}$ . 2

(c)

- (i) Use the binomial theorem to give the expansion of  $(1+x)^{2n}$ . 1

- (ii) Hence prove the following identity. 3

$${}^{2n}C_1 + 3 {}^{2n}C_3 + \dots + (2n-1) {}^{2n}C_{2n-1} = 2 {}^{2n}C_2 + 4 {}^{2n}C_4 + \dots + 2n {}^{2n}C_{2n}$$

END OF PAPER

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

Note  $\ln x = \log_e x$ ,  $x > 0$

SHORE TRIAL EXAM 2015  
Mathematics Extension 1

Section I Multiple choice

$$1. \int_0^{\frac{1}{5}} \frac{1}{\sqrt{1-25x^2}} dx = \int \frac{1}{\sqrt{25(\frac{1}{25}-x^2)}} dx$$

$$= \frac{1}{5} [\sin^{-1} 5x]_0^{\frac{1}{5}}$$

$$= \frac{1}{5} [\sin^{-1} 1 - \sin^{-1} 0]$$

$$= \frac{1}{5} [\frac{\pi}{2} - 0]$$

$$= \frac{\pi}{10}$$

(A)

$$2. 4 \log_e \sqrt{e^x} = \log(e^x)^{\frac{1}{2} \cdot 4}$$

$$= \log e^{2x}$$

$$= 2x$$

(C)

$$3. m_1 = k \quad m_2 = 2 \quad \theta = \frac{\pi}{4}$$

$$\tan \theta = \frac{|m_2 - m_1|}{1 + m_1 m_2}$$

$$\tan \frac{\pi}{4} = \frac{|2 - k|}{1 + 2k}$$

$$\pm 1 = \frac{2 - k}{1 + 2k}$$

$$2 - k = 1 + 2k \quad \text{or} \quad 2 - k = -1 - 2k$$

$$1 = 3k$$

$$k = -3$$

$$\frac{1}{3} = k$$

or

(B)

$$4. D: -1 \leq x-1 \leq 1$$

$$0 \leq x \leq 2$$

(C)

$$R: 0 \leq \frac{y}{2} < \pi$$

$$\text{or } y \leq 2\pi$$

-2-

$$5. \frac{1 - \cos 2x}{\sin 2x} = \frac{1 - (\cos^2 x - \sin^2 x)}{2 \sin x \cos x}$$

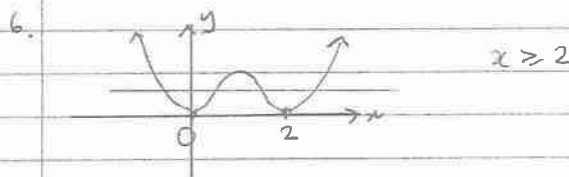
$$= \frac{(1 - \cos^2 x) + \sin^2 x}{2 \sin x \cos x}$$

$$= \frac{2 \sin^2 x}{2 \sin x \cos x}$$

$$= \frac{\sin x}{\cos x}$$

$$= \tan x$$

(D)



(A)

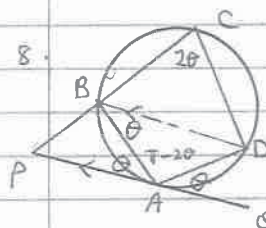
$$7. v^2 = 4(25 - x^2) \quad \text{Max speed when } x = 0$$

$$v^2 = n^2(a^2 - x^2) \quad \therefore v^2 = 4 \times 25$$

$$\therefore a = 5 \quad v = \pm 10$$

$$\text{speed} = 10 \text{ m/s}$$

(D)



(B)

$$9. \sin \frac{x}{2} = \frac{\sqrt{3}}{2}$$

$$\frac{x}{2} = n\pi \pm (-1)^n \sin^{-1} \frac{\sqrt{3}}{2}$$

$$= n\pi \pm (-1)^n \frac{\pi}{3}$$

$$x = 2n\pi \pm (-1)^n \frac{2\pi}{3}$$

(C)

10. Polynomial:  $a < 0$  y-intercept -2

A.  $-\frac{1}{6}(x+3)(x-2)^2$  y-intercept -2  $a < 0$  ✓

B.  $+\frac{1}{6}(x+3)(x+2)^2$  " " -2  $a > 0$  (A)

C.  $1(x-3)(x+2)^2$  " " -12 x

D.  $-1(x-3)(x+2)^2$  " " 12 x

## SECTION II

-3-

Marker's Comments

Question 11:

$$a) \frac{x-3}{x+4} \leq 2$$

critical values:  $x \neq -4$ 

$$\text{Solve } x-3 = 2x+8$$

$$-11 = x$$



$$\text{test } x=0 \quad -3/4 \leq 2 \quad \text{T.}$$

$$\underline{x \leq -11 \text{ or } x > -4}$$

$$\text{OR } \frac{x(x+4)^2}{x \neq -4} \leq \frac{x-3}{x+4} (x+4)^2 \leq 2(x+4)^2$$

$$(x-3)(x+4) \leq 2(x+4)^2$$

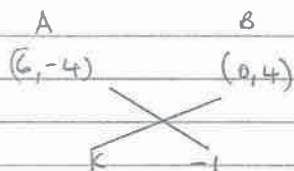
$$0 \leq 2(x+4)^2 - (x-3)(x+4)$$

$$0 \leq (x+4)(2x+8 - x+3)$$

$$0 \leq (x+4)(x+11)$$



$$b) C(-3, 8)$$



$$\therefore -3 = \frac{0-6}{k-1}$$

$$-3k+3=-6$$

$$9=3k$$

$$\underline{k=3}$$

-4-

Marker's Comments

$$c) \frac{d}{dx}(x \sin^{-1} 2x) = x \cdot \frac{2}{\sqrt{1-4x^2}} + \sin^{-1} 2x \times 1$$

$$= \frac{2x}{\sqrt{1-4x^2}} + \sin^{-1} 2x$$

$$d) \quad u = \ln x \quad x = \sqrt{e} \quad u = \frac{1}{2}$$

$$\frac{du}{dx} = \frac{1}{x} \quad x = \frac{1}{e} \quad u = \ln e^{-1}$$

$$u = -1$$

$$\int_{\frac{1}{e}}^{\sqrt{e}} \frac{dx}{x \sqrt{1-(\ln x)^2}} = \int_{-1}^{\frac{1}{2}} \frac{du}{\sqrt{1-u^2}}$$

$$= \left[ \sin^{-1} u \right]_{-1}^{\frac{1}{2}}$$

$$= \sin^{-1} \frac{1}{2} - \sin^{-1} (-1)$$

$$= \frac{\pi}{6} + \frac{\pi}{2}$$

$$= \frac{4\pi}{6}$$

$$= \underline{\underline{\frac{2\pi}{3}}}$$

$$e) y = \cos 3x$$

$$Vol = \pi \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cos^2 3x \, dx$$

$$= \pi \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \left[ \frac{1}{2} + \frac{1}{2} \cos 6x \right] dx$$

$$= \frac{\pi}{2} \left[ x + \frac{1}{6} \sin 6x \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$$

$$= \frac{\pi}{2} \left[ \left( \frac{\pi}{3} + \frac{1}{6} \sin \pi \right) - \left( \frac{\pi}{6} + \frac{1}{6} \sin \pi \right) \right]$$

$$= \frac{\pi}{2} \times \frac{\pi}{6}$$

$$= \frac{\pi^2}{12} \text{ units}^3$$

$$\begin{aligned} \text{(f)} \quad \text{let } f(x) &= 2 \ln x - \cos x \\ f(1.5) &= 2 \ln 1.5 - \cos 1.5 \\ &= 0.740193... \quad (\text{A}) \end{aligned}$$

$$f'(x) = \frac{2}{x} + \sin x$$

$$f'(1.5) = \frac{2}{1.5} + \sin 1.5$$

$$= 2.3308... \quad (\text{B})$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 1.5 - \frac{0.74...}{2.33...}$$

$$= 1.1824...$$

$$= \underline{\underline{1.2}} \quad (\text{2 sig figures})$$

## Question 12:

## Marker's Comments.

$$\text{a)} \quad \left( \frac{1}{x^2} - 2x^3 \right)^{10}$$

$$\text{General term is } {}^{10}C_k (x^{-2})^{10-k} (-2)^k (x^3)^k$$

1 mark

Constant term is the coefficient of  $x^0$ 

$$\therefore x^0 = x^{-20+2k+3k}$$

$$0 = 5k - 20$$

$$4 = k$$

1 mark

$$\therefore \text{Constant term is } {}^{10}C_4 (-2)^4 = 3360$$

1 mark

3

$$\text{(b)} \quad P(x) = x^3 + ax^2 + bx - 6 \quad \begin{array}{l} P(3) = 0 \text{ and} \\ P(-1) = 8 \end{array}$$

$$P(3) = 3^3 + a \cdot 3^2 + 3b - 6$$

$$0 = 21 + 9a + 3b \quad (+3)$$

$$\underline{-7 = 3a + b} \quad (1)$$

1 mark

$$P(-1) = -1 + a - b - 6$$

$$8 = a - b - 7$$

$$\underline{15 = a - b} \quad (2)$$

1 mark

$$(1) + (2)$$

$$8 = 4a$$

$$2 = a$$

sub into (1)

$$-7 = 6 + b$$

$$\underline{-13 = b}$$

$$\therefore \underline{\underline{a=2 \quad b=-13}}$$

1 mark

3



Question 13:

(a)  $T = 22 + Be^{-kt}$

(i)  $t=0$   $T=180$

$$180 = 22 + Be^0$$

$$158 = B$$

$$T = 22 + 158e^{-kt}$$

When  $t=10$   $T=115$   $115 = 22 + 158e^{-10k}$

$$\frac{93}{158} = e^{-10k}$$

$$k = \frac{\ln \frac{93}{158}}{-10}$$

$$= -\frac{1}{10} \ln \frac{93}{158}$$

(ii)  $T=23$   $23 = 22 + 158e^{-kt}$

$$\frac{1}{158} = e^{-kt}$$

$$-\frac{1}{k} \ln \frac{1}{158} = t$$

$$t = 95.52 \text{ minutes}$$

$$\text{time} = 11:30 + 1 \text{ hr } 36'$$

$$= \underline{1:06 \text{ pm}}$$

Marker's Comments

Marker's Comments

(b)  $\ddot{x} = -e^{-\frac{x}{2}}$

$$t=0 \quad x=0 \quad v=2$$

(i)  $\frac{d}{dx}(\frac{1}{2}v^2) = -e^{-\frac{x}{2}}$   
 $\frac{1}{2}v^2 = -\frac{e^{-\frac{x}{2}}}{-\frac{1}{2}} + c$

$$= 2e^{-\frac{x}{2}} + c$$

$$2 = 2e^0 + c$$

$$0 = c$$

$$\left. \begin{matrix} v=2 \\ x=0 \end{matrix} \right\}$$

$$\therefore \frac{1}{2}v^2 = 2e^{-x/2}$$

$$v^2 = 4e^{-x/2}$$

(ii)  $v = \sqrt{4e^{-x/2}} \quad v > 0$   
 $= 2e^{-x/4}$

$$\frac{dx}{dt} = \frac{2}{e^{x/4}}$$

$$\frac{dt}{dx} = \frac{e^{x/4}}{2}$$

$$t = \frac{1}{2} \frac{e^{x/4}}{\frac{1}{4}} + c$$

$$= 2e^{x/4} + c$$

$$\left. \begin{matrix} t=0 \\ x=0 \end{matrix} \right\}$$

$$0 = 2e^0 + c$$

$$-2 = c$$

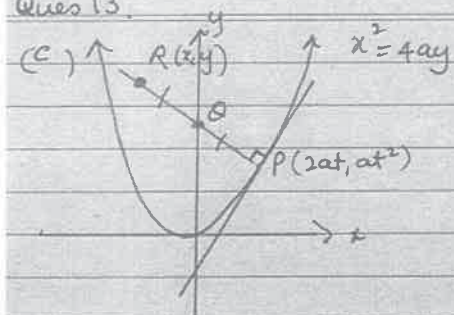
$$t = 2e^{x/4} - 2$$

$$\frac{t+2}{2} = e^{x/4}$$

$$\ln\left(\frac{t+2}{2}\right) = \frac{x}{4}$$

$$\underline{4 \ln\left(\frac{t+2}{2}\right) = x}$$

Ques 13.



(i)  $y = \frac{x^2}{4a}$   
 $\frac{dy}{dx} = \frac{2x}{4a}$

At P  $x = 2at$   $\frac{dy}{dx} = \frac{4at}{4a} = t$

$m_T = t \therefore m_N = -\frac{1}{t}$

Eqn of normal:  $y - at^2 = -\frac{1}{t}(x - 2at)$   
 $yt - at^3 = -x + 2at$   
 $x + ty = 2at + at^3$

(ii) let  $x = 0$   $ty = 2at + at^3$   
 $y = 2a + at^2$   
 $= a(2 + t^2)$

$Q(0, a(2 + t^2))$  (midpt of RP)

$\therefore R: -0 = \frac{x + 2at}{2}$   $a(2 + t^2) = \frac{y + at^2}{2}$   
 $-2at = x$

$2a(2 + t^2) = y + at^2$   
 $-at^2 + 4a + 2at^2 = y$   
 $y = at^2 + 4a$

$\therefore R(-2at, at^2 + 4a)$

Marker's Comments

13.

(ii) Locus of R.

$x = -2at$   $y = at^2 + 4a$  (2)  
 $\frac{x}{-2a} = t$  (1)

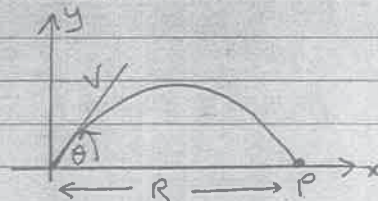
Sub (1) into (2)

$y = a\left(\frac{x}{-2a}\right)^2 + 4a$   
 $= \frac{ax^2}{4a^2} + 4a$

$4a(y - 4a) = x^2$   
 $x^2 = 4a(y - 4a)$

which is a parabola with vertex  
 $(0, 4a)$

Question 14.



(i)  $x = vt \cos \frac{\pi}{4}$   $y = vt \sin \frac{\pi}{4} - \frac{1}{2}gt^2$   
 $x = \frac{vt}{\sqrt{2}}$   $= \frac{vt}{\sqrt{2}} - \frac{1}{2}gt^2$  (2)

$\frac{\sqrt{2}x}{v} = t$  (1) ✓

Sub (1) into (2)

$y = \frac{v}{\sqrt{2}}\left(\frac{\sqrt{2}x}{v}\right) - \frac{1}{2}g\left(\frac{\sqrt{2}x}{v}\right)^2$

$y = x - \frac{1}{2}g \cdot \frac{2x^2}{v^2}$

$y = x - \frac{gx^2}{v^2}$  ✓

Well done

1 mark for t

1 mark for correct  
 substitution (which  
 must be shown)

(ii) To find R, let  $y=0$  find  $x$

$$0 = x - g \left( \frac{x^2}{v^2} \right)$$

$$0 = x - g \frac{x^2}{v^2}$$

$$0 = x \left( 1 - \frac{gx}{v^2} \right)$$

$$\therefore x=0 \text{ or } 1 - \frac{gx}{v^2} = 0$$

(origin)  $1 = \frac{gx}{v^2}$

$$\frac{x^2}{g} = x \quad \text{but } x=R$$

$$\text{ie } R = \frac{v^2}{g}$$

(iii)  $V=30$   $x=15$ .

$$x = vt \cos \theta$$

$$15 = \frac{30t}{\sqrt{2}}$$

$$\frac{15\sqrt{2}}{30} = t$$

$$t = \frac{\sqrt{2}}{2} \text{ s}$$

$$\dot{x} = \frac{V}{\sqrt{2}} = \frac{30}{\sqrt{2}} = \frac{15\sqrt{2}}{1}$$

$$At = \frac{\sqrt{2}}{2} \quad \dot{y} = \frac{30}{\sqrt{2}} - 5\sqrt{2}$$

$$= \frac{15\sqrt{2}}{1} - 5\sqrt{2}$$

$$= 10\sqrt{2}$$

$$\dot{y} = \frac{V}{\sqrt{2}} - gt$$

$$= \frac{30}{\sqrt{2}} - 10t$$

$$V = \sqrt{\dot{x}^2 + \dot{y}^2}$$

$$= \sqrt{900 + 200}$$

$$= \sqrt{1100}$$

$$= \sqrt{650}$$

$$V = 25.5 \text{ m/s}$$

Speed = 25.5 m/s.

Marker's Comments

Well done.

$\frac{1}{2}$  mark

$\frac{1}{2}$  mark

Students must  
factorise  $x$ , not  $\frac{1}{2}x$

1 mark.

[2]

Students who found  
t "usually were  
successful in  
finding the speed.

$\frac{1}{2}$  mark for t.

$\frac{1}{2}$  mark  $\dot{x}$  } at  $t = \frac{\sqrt{2}}{2}$   
 $\frac{1}{2}$  mark  $\dot{y}$

$\frac{1}{2}$  mark.

$\frac{1}{2}$  mark.

$\frac{1}{2}$  mark correct  
calculation [3]

14.

(b)



$$\frac{dx}{dt} = 2 \text{ cm/s}$$

$$dt = 0.02 \text{ m/s}$$

$$\tan \theta = \frac{2.4}{x}$$

$$x = \frac{2.4}{\tan \theta}$$

$$= 2.4 (\tan \theta)^{-1}$$

$$\frac{dx}{d\theta} = -2.4 (\tan \theta)^{-2} \sec^2 \theta$$

$$= -\frac{2.4 \sec^2 \theta}{\tan^2 \theta}$$

$$= -\frac{2.4 \cdot \frac{\cos^2 \theta}{\sin^2 \theta} \cdot \sin^2 \theta}{\sin^2 \theta}$$

$$= -\frac{2.4}{\sin^2 \theta}$$

Marker's Comments

$\frac{1}{2}$  mark.

1 mark

$\frac{1}{2}$  mark Correct simplified

OR.

$$y = \frac{2.4 \cos \theta}{\sin \theta}$$

$$u = 2.4 \cos \theta$$

$$u' = -2.4 \sin \theta$$

$$v = \sin \theta$$

$$v' = \cos \theta$$

$$\frac{dx}{d\theta} = \frac{vu' - uv'}{v^2}$$

$$= \frac{\sin \theta \cdot -2.4 \sin \theta - 2.4 \cos \theta \cos \theta}{\sin^2 \theta}$$

$$= -\frac{2.4 (\sin^2 \theta + \cos^2 \theta)}{\sin^2 \theta}$$

$$= -\frac{2.4}{\sin^2 \theta} \quad \text{which is required}$$

|                                                                                                                                                                                                                                                                                                  | Marker's comments                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(ii) <math>\frac{d\theta}{dt} = \frac{d\theta}{dx} \cdot \frac{dx}{dt}</math></p> <p><math>= \frac{\sin^2 \theta}{-2.4} \times 0.02</math></p> <p><math>= \frac{-\left(\frac{1}{\sqrt{2}}\right)^2 \times 0.02}{-2.4}</math></p> <p><math>= \underline{\underline{-\frac{1}{240}}}</math></p> | <p><math>\frac{1}{2}</math> units (cm = 0.02m)</p> <p><math>\frac{1}{2}</math> substitution into correct product</p> <p>correct answer.</p> |
| <p><math>\therefore</math> angle is decreasing by <math>\frac{1}{240}</math> radians/s</p>                                                                                                                                                                                                       | <p>2</p>                                                                                                                                    |
| <p>(c) (i) <math>(1+x)^{2n} = {}^{2n}C_0 + {}^{2n}C_1 x + {}^{2n}C_2 x^2 + \dots + {}^{2n}C_n x^{2n}</math></p>                                                                                                                                                                                  | <p>1 mark. 1</p>                                                                                                                            |
| <p>(ii) Differentiate b.s (i) wrt x</p> <p><math>2n(1+x)^{2n-1} = {}^{2n}C_1 + 2 {}^{2n}C_2 x + 3 {}^{2n}C_3 x^2 + \dots + k {}^{2n}C_k x^k</math><br/> <math>+ \dots + 2n {}^{2n}C_{2n} x^{2n-1}</math></p>                                                                                     | <p>1 mark</p>                                                                                                                               |
| <p>Substitute <math>x = -1</math></p> <p><math>0 = {}^{2n}C_1 - 2 {}^{2n}C_2 + 3 {}^{2n}C_3 - 4 {}^{2n}C_4 + \dots + (-1)^n 2n {}^{2n}C_{2n}</math></p>                                                                                                                                          | <p>1 mark</p>                                                                                                                               |
| <p>Move the negative terms to LHS of the eqn</p> <p><math>2 {}^{2n}C_2 + 4 {}^{2n}C_4 + 6 {}^{2n}C_6 + \dots + 2n {}^{2n}C_{2n}</math><br/> <math>= {}^{2n}C_1 + 3 {}^{2n}C_3 + 5 {}^{2n}C_5 + \dots + (2n-1) {}^{2n}C_{2n-1}</math></p>                                                         | <p>1 mark</p> <p>3</p>                                                                                                                      |